

Silent Undertreatment: Does Racial Pain-Treatment Bias Persist in Language Models?

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Team Responsibilities:

- Sofia: Conduct data exploration (look at data distribution for each feature, etc.)
- Yanfei: Pre-process data (removing data points with empty cells, feature engineering pain level)
- Swati: Develop model pipeline (using Gemini API) and develop final graphs/charts after prompting
- Everyone:
 - Contribute to writing the paper
 - Manually labelling model output

1. Research Thesis and Course Connections

Central Question: Does an existing LLM provide different pain management approaches for a patient depending on their race and gender? This matters because LLMs are now being used for clinical decision support, documentation, and communication with patients. If the model reproduces the real-world bias of under-treating under-represented patients, it may automate and scale an existing healthcare inequity as opposed to correcting it.

Field Landscape:

- Representation bias - see if the LLM model is having different approaches for pain management for people based on race and gender is their representation for under-represented groups in data?
 - Suresh, Harini, and John Gutttag. "A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle." *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 2021, pp. 1–9. arXiv, <https://doi.org/10.48550/arXiv.1901.10002>.
- Intersectional accuracy disparities - There are features and information about people that differ based on their gender and/or race; this could then affect ML models and how they interact with datasets regarding minorities
 - Buolamwini, J. & Gebru, T.. (2018). "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification." *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, in Proceedings of Machine Learning Research 81:77-91 Available from <https://proceedings.mlr.press/v81/buolamwini18a.html>.

Academic Contribution: This is showing that if there is bias within the dataset, the model developer using it to train an AI model will still make biased decisions, thus perpetuating discrimination already existing in healthcare.

2. Literature and Theoretical Framework (reference list is not included in the 1 page)

Project-Specific Sources: We will be using the [Google Gemini API documentation](#):

External Sources:

- Hoffman KM, Trawalter S, Axt JR, Oliver MN. Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites. *Proc Natl Acad Sci U S A*. 2016 Apr 19;113(16):4296-301. doi: 10.1073/pnas.1516047113. Epub 2016 Apr 4. PMID: 27044069; PMCID: PMC4843483.
 - Discusses how Black patients are systematically undertreated for pain

- Omiye, J.A., Lester, J.C., Spichak, S. et al. Large language models propagate race-based medicine. npj Digit. Med. 6, 195 (2023). <https://doi.org/10.1038/s41746-023-00939-z>
 - Tests different prompts relating to healthcare on ChatGPT, Claude, GPT-4, and Bard - found that there were very concerning outputs when patient's race was involved and wasn't white
- Zack T, Lehman E, Suzgun M, Rodriguez JA, Celi LA, Gichoya J, Jurafsky D, Szolovits P, Bates DW, Abdulnour RE, Butte AJ, Alsentzer E. Assessing the potential of GPT-4 to perpetuate racial and gender biases in health care: a model evaluation study. Lancet Digit Health. 2024 Jan;6(1):e12-e22. doi: 10.1016/S2589-7500(23)00225-X. Erratum in: Lancet Digit Health. 2024 Jul;6(7):e445. doi: 10.1016/S2589-7500(24)00120-1. PMID: 38123252.
 - Found there's a need for comprehensive and transparent bias assessments of LLM tools such as GPT-4

Theoretical Approach: The framework we are using to guide our research is algorithmic fairness in LLM models to see if it will provide the same suggestions for patients at the same level of pain despite their race and gender.

Gap Analysis: We will examine if racial and gender biases affect pain-management recommendations when only patients' race and gender differ.

Integration: We will be building different personas with different representations of people and see what outputs it provides. How LLM is suggesting different approaches for patients of different race and/or gender. We are showing bias in a clinical context using LLM prompts using a diabetic patient dataset

3. Methodology and Research Design

Model: Google Gemini 3.8 Flash

Dataset: [Diabetes 130-US Hospitals for Years 1999-2008 from UCI](#)

Metric: We will prompt the LLM model based on multiple personas and will see how refusal rate and recommendation strength varies

Comparison groups: Black patient, White patient, Hispanic patients (and will do gender subgroups as well)

Sample size: 50 prompts

Output:

- CSV file containing pain management labels
- One table and one chart showing the refusal rates between the different groups and subgroups
- A paragraph showing if this model continues to perpetuate health inequities

Fits the bar: Audits one LLM using controlled demographic prompts and quantitative comparisons.

Doesn't fit the bar: Does not train or fine-tune models, build a new dataset, conduct user studies, or audit multiple systems.

4. Ethical and Societal Implications

Potential risks and stakeholder impacts, informed by your audit study's findings.

- If disparities are found, then it may not be best to use this LLM model since the bias it perpetuates would affect real pain-treatment. Also, decisions should not be made solely off of the LLMs output; there should be a human in the loop.
- Stakeholder (if AI is deployed) - clinicians (may over-rely on AI), patients (may not receive equal care)

Post-deployment considerations for real implementation: Being transparent about data used and having a human in the loop to make final decision

How you'll address bias, fairness, transparency, and accountability: Show discrepancy for approaches for under-represented groups if they come with the same pain level

5. Open Questions and Execution Risks

- Risk - Inaccurately labeling output based on our judgement
- Sample misfit- Will only be using a small sample so may not be representative of the larger section of data
- Execution risk - Don't have any pain management specific dataset, so we are data engineering a feature from this existing Diabetics dataset to represent pain level so are unsure if the gap we are looking for will actually be presented.